

Convergent retention of oxidative-defense orthogroups across independent extremotolerant origins: A comparative-proteomics screen in Deinococcaceae, Tardigrada, and Bdelloidea

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1 Introduction (skeleton)

Extremotolerant organisms have evolved radiation resistance independently at least three times: in the bacterial family Deinococcaceae, in tardigrades, and in bdelloid rotifers. Whether these origins converged on the same molecular machinery is unknown. We test whether orthogroups are convergently retained across resistant lineages while absent from radiation-sensitive controls.

2 Background: The Biology of Extreme Radiation Resistance

2.1 Ionizing radiation as a selective force

Ionizing radiation damages cells through two routes. Direct deposition of energy in DNA produces double-strand breaks (DSBs), the most lethal common lesion; a single unrepaired DSB can be sufficient to kill a dividing cell. Indirect damage is mediated by water radiolysis, which generates hydroxyl radicals, superoxide, and hydrogen peroxide, and these reactive oxygen species (ROS) oxidize proteins, lipids, and nucleic acids far from the site of the original photon absorption. For desiccation-tolerant organisms the indirect route dominates: at low water activity the primary lesion burden shifts toward oxidative protein carbonylation, and survival correlates more strongly with proteome protection than with DNA repair capacity alone.

2.2 The canonical model: *Deinococcus radiodurans*

D. radiodurans R1 survives acute gamma doses above 10 kGy without loss of viability, roughly three orders of magnitude beyond the lethal dose for *Escherichia coli*. Three decades of work have established that its resistance is not attributable to unusual target hardening of DNA. Instead, a layered defense acts on the proteome and the metabolome: (i) manganese accumulation, with intracellular Mn(II) concentrations reaching millimolar levels and Mn/Fe ratios far above those of sensitive bacteria; (ii) Mn(II) complexes with orthophosphate and small metabolites (peptides, nucleosides) that scavenge superoxide and hydroxyl radicals non-enzymatically; (iii) an unusually deep enzymatic antioxidant battery, including multiple catalases, superoxide

dismutases, and peroxiredoxins of the AhpC/TSA family; and (iv) an efficient homologous-recombination machinery operating on a multipartite, polyploid genome that reassembles shattered chromosomes from overlapping fragments through extended synthesis-dependent strand annealing. The resulting picture is that of a cell whose DNA is damaged at the same rate as any other cell's but whose proteins remain functional throughout irradiation, permitting rapid repair once conditions allow.

2.3 Convergent resistance in other clades

Extreme resistance has evolved independently in several lineages, and each case is an independent natural experiment. Within bacteria, the Deinococcaceae contain multiple resistant species (*D. radiodurans*, *D. geothermalis*, *D. deserti*), and the family is treated in this study as a single origin because of shared ancestry. Among animals, tardigrades such as *Ramazzottius varieornatus* tolerate kilogray doses in the desiccated state, and at least one lineage-specific protein, Dsup, binds nucleosomes and reduces DNA damage in heterologous systems, including human cells. Bdelloid rotifers such as *Adineta vaga* combine desiccation tolerance with radiation resistance and carry heavily reorganized genomes with signatures of horizontal gene transfer, including genes of bacterial and fungal origin implicated in stress response. These three origins - Deinococcaceae, Tardigrada, and Bdelloidea - are separated by more than a billion years of evolution, so any molecular feature shared across them is a candidate convergence on a common solution rather than a shared inheritance.

2.4 Why oxidative defense is the leading axis

Across resistant taxa the recurring observation is elevated antioxidant capacity: peroxiredoxins, glutaredoxin and thioredoxin systems, flavin reductases that recycle redox cofactors, and metal-homeostasis systems that limit iron-catalyzed Fenton chemistry. Peroxiredoxins of the AhpC/TSA family are peroxide-scavenging enzymes with exceptionally high catalytic rates; their up-regulation is a common response to irradiation in bacteria, and their abundance correlates with desiccation tolerance in tardigrades and rotifers. If radiation resistance across kingdoms shares a single dominant mechanistic axis, comparative genomics should recover orthogroups in this pathway as convergently retained or expanded in resistant lineages while absent from sensitive relatives - the hypothesis tested here.

2.5 The specificity problem

A central confound is that desiccation, oxidation, heat, and radiation stresses are correlated in nature. Organisms that survive drying necessarily solve the oxidative-damage problem, and radiation resistance may in part be a byproduct of desiccation adaptation. Any genomic enrichment found in resistant species could therefore reflect general stress adaptation rather than radiation-specific selection. This study addresses the confound directly with a pre-registered, stress-matched control panel (Section ??): stress-tolerant but radiation-sensitive species are subjected to the identical pipeline, and candidate orthogroups must be absent from that panel to retain a radiation-specific interpretation.

2.6 Known mechanisms of ionizing-radiation resistance

Decades of work on *Deinococcus radiodurans* established that extreme resistance is not conferred by a single pathway. Three mechanistic layers are recognized. First, an exceptionally clean antioxidant defense: high cellular Mn^{2+}/Fe^{2+} ratios, manganese-phosphate metabolite complexes, and small-molecule antioxidants protect proteins — and especially the DNA-repair enzymes themselves — from oxidative inactivation, so that an otherwise conventional repair toolkit keeps working at doses that disable it in sensitive cells. Second, efficient and redundant DNA-repair capacity: extended synthesis-dependent strand annealing, RecA-dependent homologous recombination, and non-homologous end-joining reassemble chromosomes shattered into hundreds of fragments. Third, genome organization and proteome protection: nucleoid condensation, redundant genome copies, and chaperone/protease systems limit and clear damaged protein. Tardigrades add distinct intrinsically disordered protective proteins (CAHS, SAHS, and the DNA-shielding Dsup), demonstrating that more than one molecular route to resistance exists; bdelloid rotifers combine desiccation-linked antioxidant defense with horizontal-gene-transfer-enriched damage-response repertoires. A comparative screen across independent origins therefore asks not whether a single known factor is shared — none is — but which *modules* were independently retained, which is the question the locked gates operationalize.

2.7 Why convergence is the informative signal

Single-organism studies cannot separate general stress biology from radiation-specific adaptation, and single-lineage comparative genomics cannot separate ancestry from adaptation. Convergent retention across independent extremotolerant origins — here a bacterial family and a metazoan class — restricts attention to modules that selection kept rebuilding or re-keeping under the same pressure, while the stress-matched radiation-sensitive control panel removes modules that merely mark hardy organisms in general. This design logic follows the comparative strategy that identified convergent desiccation machinery, applied here to the radiation axis with pre-registered gates so that every retained or discarded candidate carries a stated reason.

3 Methods (pre-registered; amendments dated 2026-09-26)

Panel: 5 resistant proteomes available of the 6 locked (*Deinococcus radiodurans*, *D. geothermalis*, *D. deserti* [origin 1: *Deinococcaceae*]; *Ramazzottius varieornatus* [origin 2: *Tardigrada*]; *Adineta vaga* [origin 3: *Bdelloidea*]) vs 6 sensitive controls (*Thermus aquaticus*, *T. thermophilus*, *Hypsibius exemplaris*, *Caenorhabditis elegans*, *Drosophila melanogaster*, *Escherichia coli*). *Milnesium tardigradum* has no reference assembly; *Tardigrada* origin is single-proteome (power caveat carried throughout).

Orthogroup calling: MMseqs2 v17 sensitive clustering (min_seq_id 0.3, coverage 0.5, cov-mode 1), 154,665 proteins. Linclust used only as sensitivity; subset concordance (20k proteins): linclust pair purity 0.914 vs sensitive, retention 0.363, so all formal claims rest on sensitive clustering.

Formal test (locked 11:15 and 11:29 amendments, before outcome inspection): per orthogroup, one-sided Fisher exact on presence/absence (5 resistant vs 6 sensitive; the sixth locked resistant species, *Milnesium tardigradum*, has no reference assembly) and one-sided Mann-Whitney U on zero-filled copy numbers. The BH-FDR gate was replaced pre-outcome by 1000 label permutations with per-orthogroup empirical p , after W03's power analysis showed BH-FDR is structurally impossible at this panel size (minimum achievable Fisher $p = 1/\binom{11}{5} =$

2.16×10^{-3} at the true 5-vs-6 panel vs BH rank-1 threshold $\approx 10^{-6}$). Gate: empirical $p < 0.05$, resistant-direction, replication in ≥ 2 of 3 independent resistant origins.

Candidate annotation: NCBI esummary + hmmscan (HMMER 3.3.2) vs Pfam-A (full-sequence $E < 10^{-10}$).

3.1 Formal framework

Each orthogroup g is a partition of the panel proteomes into member sets; for species i let $n_{gi} \geq 0$ be the copy number and $x_{gi} = \mathbf{1}[n_{gi} > 0]$ the presence indicator. The resistant and sensitive label vectors $y \in \{0, 1\}^{11}$ are fixed by the panel definition. The presence/absence arm applies a one-sided Fisher exact test to the 2×2 table of x_g against y ; under complete separation (all resistant present, all sensitive absent) the minimum achievable value is

$$p_{\min} = \frac{1}{\binom{11}{5}} = 2.16 \times 10^{-3}, \quad (1)$$

which motivated replacing the BH-FDR gate by permutation calibration (pre-outcome amendment). The empirical p -value for orthogroup g over $B = 1000$ label permutations π_b is

$$\hat{p}_g = \frac{1 + \sum_{b=1}^B \mathbf{1}[T(x_g, \pi_b(y)) \geq T(x_g, y)]}{1 + B}, \quad (2)$$

with statistic T the resistant-direction enrichment score; the empirical- p estimator floor is $1/(1+B) = 1/1001 \approx 10^{-3}$; the observed $\hat{p} = 0.013$ (13/1001) sits 13 permutation counts above that floor and is a genuine count-based estimate, not a floor-limited value. The copy-number arm uses a one-sided Mann-Whitney statistic

$$U_g = \sum_{i: y_i=1} \sum_{j: y_j=0} \mathbf{1}[n_{gi} > n_{gj}] + \frac{1}{2} \mathbf{1}[n_{gi} = n_{gj}] \quad (3)$$

on zero-filled copy numbers. The convergence gate additionally requires replication across independent origins:

$$\sum_{k \in \{\text{Deino, Tardi, Bdello}\}} \mathbf{1} \left[\max_{i \in k} x_{gi} = 1 \right] \geq 2. \quad (4)$$

For the specificity layer, each v1 candidate was re-clustered with four stress-tolerant but radiation-sensitive controls and re-tested identically; a candidate keeps the label radiation-specific only if

$$|\{c \in \text{stress controls} : x_{gc} = 1\}| \leq 1, \quad (5)$$

otherwise it is re-labeled broad-stress adaptation. Domain-level confirmation searches each candidate's full Pfam architecture $\mathcal{D}_g = \{d_1, \dots, d_m\}$ against the five confirmation proteomes with family HMMs, accepting a proteome as supporting iff

$$\exists \text{protein } q : \bigwedge_{d \in \mathcal{D}_g} [E(q, d) \leq 10^{-5}], \quad (6)$$

and confirming the candidate iff at least two of five proteomes support it. The phylogenetic layer (H4–H4c) maps presence onto a fixed cladogram τ and counts minimum gains by Fitch reconstruction,

$$N_{\text{ind}}(g) = \min_{\text{ASR}} \sum_{e \in \tau} \mathbf{1}[0 \rightarrow 1 \text{ on } e, e \text{ subtends a resistant leaf}], \quad (7)$$

against a null of 10^4 within-strata presence shuffles σ_r preserving stratum counts,

$$p_{\text{lower}}(g) = \frac{1 + \sum_r \mathbf{1}[N_{\text{ind}}^{\sigma_r} \leq N_{\text{ind}}(g)]}{1 + 10^4}, \quad (8)$$

which proved degenerate on saturated strata (zero null variance), yielding the identifiability boundary reported in Section 3.4.

3.2 Pipeline definition (end-to-end operator form; amendment 2026-09-27 12:08 IST)

Collected, the framework is a single operator from raw inputs to a decision-relevant verdict. The inputs are the panel proteomes $\mathcal{P} = \{P_1, \dots, P_{20}\}$, the fixed resistant/sensitive label vector y , and the background orthogroup universe $\mathcal{M}$ ($|\mathcal{M}| = 52,786$). The pipeline is the composition

$$\Phi(\mathcal{P}, y, \mathcal{M}) = A \circ D \circ S \circ E \circ C(\mathcal{P}), \quad (9)$$

where C clusters proteomes into orthogroups; E is the enrichment gate, $E(g) = \mathbf{1}[\hat{p}_g < \alpha_{\text{emp}}] \wedge \mathbf{1}[\text{origins}(g) \geq 2]$ (empirical permutation p and independent-origin replication, equations above); S is the stress-control specificity filter; D is the Pfam-architecture domain confirmation against independent proteomes; and A is the phylogenetic identifiability audit of Section 4.5, which returns the class of equation (13) for every candidate, candidate or not. The output is *not* a scalar score: each candidate receives the verdict tuple

$$v(g) = (\text{label}(g), \text{evidence tier}(g), \text{class}(g)), \quad (10)$$

mapped onto the ordered confidence scale

$$\begin{aligned} \text{CONFIRMED-CONVERGENT} &> \text{CANDIDATE-TESTABLE} > \text{CANDIDATE-UNDERPOWERED} \\ &> \text{CANDIDATE-UNTESTABLE} > \text{REJECTED}, \end{aligned} \quad (11)$$

where the first tier requires $\text{class}(g) = \text{TESTABLE}$ *and* a significant convergence statistic, and a scalar collapse is deliberately refused: on any panel where the audit returns IMPOSSIBLE or UNDERPOWERED, a numeric “confidence” would manufacture precision the data cannot support. The identifiability class is therefore a *precondition* for interpreting every other stage’s output: enrichment, specificity, and domain confirmation establish that a candidate is real and radiation-associated; only the audit establishes whether the panel could ever have told us otherwise. This is the framework’s stated purpose - a pipeline that prevents false convergence, including against itself: applied to its own output (Table 2), it returns 9 IMPOSSIBLE and 1 UNDERPOWERED, and no candidate is reported as convergent.

4 Results

Of 52,786 orthogroups tested, **12 pass all locked gates**: convergently present in Deinococcaceae and Bdelloidea, absent from all 6 sensitive controls, empirical $p = 0.013$ (1000 permutations). A further 26 orthogroups pass the copy-number Mann-Whitney arm.

The annotated candidates concentrate on the oxidative-defense axis: peroxiredoxin (AhpC-TSA), thioredoxin-family oxidoreductase (DSBA), and flavin reductase. Ionizing-radiation damage is dominated by reactive oxygen species; convergent retention of an oxidative-defense

Table 1: Convergent orthogroups passing all locked gates, with Pfam domain calls.

Representative	Pfam domains (E-value)	Interpretation
A. vaga UJR09360.1	AhpC-TSA (3.8e-35), Redoxin (8.8e-17)	Peroxiredoxin; peroxide detox
D. geothermalis WP_272975990.1	DSBA (7e-50)	Thiol-disulfide oxidoreductase
D. deserti WP_012692142.1	Flavin_Reduct (8.1e-26)	Flavin reductase; redox
D. deserti WP_012692117.1	Ank $\times$ 5 (6e-24...)	Ankyrin scaffold; DDR interactions
D. geothermalis WP_414646549.1	PAD_porph (7.7e-122)	Porphyrin/heme binding
D. geothermalis WP_272975981.1	Acetyltransf_1/7	GNAT acetyltransferase
A. vaga UJR10666.1	HPPK (3.3e-29), Pterin_bind	Folate/pterin pathway
A. vaga UJR11615.1	DHH (2.8e-21), DHHA2 (2e-22)	Phosphoesterase; nucleotide damage
A. vaga UJR26281.1	HAD_2 (2.2e-25), Hydrolase	HAD-family hydrolase
A. vaga UJR28130.1	GDP_Man_Dehyd, NmrA	Redox-sensing dehydrogenase
A. vaga UJR36058.1	DUF4385 (2.6e-60)	Uncharacterized (novel space)
A. vaga UJR17759.1	Amidase (2.8e-61), Bac_rhamnosid6H	Cell-wall/envelope metabolism

module across a bacterial and a metazoan origin, with complete absence from sensitive controls, is the screen’s central signal. The folate/pterin candidate (HPPK) is consistent with the known *Deinococcus* folate-photoprotection mechanism, and DHH/DHHA2 phosphoesterases act in nucleotide-damage processing.

4.1 Specificity layer on the extended panel (amendment 2026-09-26, 13:18 IST)

The panel was extended pre-data with four stress-tolerant but radiation-*sensitive* controls — *Pyrococcus furiosus*, *Sulfolobus solfataricus*, *Lactobacillus plantarum*, and *Saccharomyces cerevisiae* — chosen to separate radiation-specific retention from generic stress adaptation. All 168,186 proteins from the 15 proteomes were re-clustered de novo at the locked parameters (min. seq. id. 0.3, coverage 0.5, cov-mode 1), yielding 60,164 clusters. The locked gates were re-run identically on this v2 clustering, and each v1 candidate was then required to be absent from at least 3 of the 4 new controls to retain the label *radiation-specific*; otherwise it is re-labeled *broad-stress adaptation*.

All 12 v1 representative proteins mapped into v2 clusters. **10 of 12 candidates pass both the re-run enrichment gate (empirical $p = 0.013$ – 0.017 , 1000 permutations) and the specificity layer**, each with *Deinococcaceae*+*Bdelloidea* origins and at most one control-species member. Two honest negatives: (i) the v1 lead, the AhpC-TSA peroxiredoxin (UJR09360.1), is present in 3 of 4 stress-tolerant controls and is re-labeled broad-stress adaptation — biologically consistent, since peroxiredoxins are near-universal stress proteins; (ii) the HPPK/pterin candidate (UJR10666.1) lost its *Deinococcaceae* members in the larger panel (*Bdelloid*-only cluster, empirical $p = 0.449$) and no longer shows cross-kingdom convergence.

The oxidative-defense axis survives the specificity layer through the DSBA thiol-disulfide oxidoreductase and the flavin reductase candidates, now supported as radiation-specific rather than generic stress adaptations. The surviving set additionally includes the ankyrin scaffold, PAD-porphyrin binding protein, GNAT acetyltransferase, DHH/DHHA2 phosphoesterase, HAD hydrolase, GDP-mannose dehydrogenase, amidase, and the uncharacterized DUF4385 protein — the last representing genuinely novel candidate space.

4.2 Domain-level orthogonal confirmation (amendment 2026-09-26, 20:21 IST)

As an independent check, the panel was extended with five radiation-resistant proteomes never used in the screen: *Deinococcus radiophilus*, *D. proteolyticus*, *D. maricopenensis*, *Adineta ricciae*, and *Rotaria magnacalcarata* (95,758 proteins; accessions and hashes in the ledger). For each radiation-specific candidate, its complete v1 Pfam domain set was searched against all five new proteomes (64 family HMMs; domain calls at $E \leq 10^{-5}$, fixed before candidate-level results). **9 of 10 candidates are confirmed:** their full domain architectures recur in at least one protein of at least two of the five new proteomes (eight at 5/5 support, one at 4/5). The five-domain amidase architecture (UJR17759.1) found no complete match and is labeled “domain-level support absent in extended panel” without removal from the candidate set (locked rule); the HPPK candidate that failed v2 enrichment is likewise unconfirmed, a consistent picture.

4.3 Phylogenetically controlled convergence test (amendments H4–H4c, 2026-09-26)

The strongest remaining objection to the v2 labels is phylogenetic confounding: the three *Deinococcus* proteomes (like the two *Thermus* controls) are not independent replicates, so a label permutation that treats species as exchangeable only partially randomizes family-level structure. We therefore implemented a phylogenetically controlled convergence test, locked before computation (amendment H4): for each radiation-specific candidate, presence/absence was mapped onto a fixed 15-species cladogram (topology from NCBI taxonomy, unit branch lengths), and the minimum number of independent gains into resistant-subtending branches was computed by Fitch-parsimony ancestral-state reconstruction (the minimum-gain count is method-independent; a PastML maximum-likelihood cross-check is queued). The null was 10,000 structure-preserving permutations that shuffle presence assignments only within exchangeable strata (Deinococcaceae, *Thermus*, other controls), never across. Formally, let $x_g \in \{0, 1\}^{|\text{leaves}(\tau)|}$ be the presence vector of candidate g on the fixed cladogram τ . The Fitch bottom-up pass assigns each internal node v the intersection of its children’s state sets when non-empty, else their union, and the minimum-gain statistic is the number of union events,

$$G(x_g; \tau) = \sum_{v \in \text{int}(\tau)} \mathbf{1}[S_v = S_{v_1} \cup S_{v_2}], \quad (12)$$

where S_v is the Fitch state set of node v with children v_1, v_2 , and a union event implies exactly one gain on the branch subtending the child absent the state. The structure-preserving null draws permutations π_b that shuffle presence labels only within exchangeable strata $\mathcal{S} \in \{\text{Deinococcaceae}, \text{Thermus}, \text{other controls}\}$, and the corrected (lower-tail, amendment H4c) empirical convergence p -value is

$$\hat{p}_g^{\text{conv}} = \frac{1 + \sum_{b=1}^B \mathbf{1}[G(\pi_b(x_g); \tau) \leq G(x_g; \tau)]}{1 + B}, \quad (13)$$

with $B = 10,000$; the lower tail is the convergence-consistent direction because scattered presence forces *more* gains, so unusually *few* gains reject the structured null.

The test returned a structural null: for all ten candidates the null distribution has zero variance (observed 2 gains, null median 2, $p = 1.0$). The cause is identifiability, not biology: every candidate is present in all three Deinococcaceae, and the Tardigrada and Bdelloidea origins are singletons whose labels cannot be permuted, so no within-strata shuffle can move the signal. Resistance labels are perfectly confounded with tree topology on this panel.

We then locked an extended-panel arm (amendment H4b) adding the five confirmation lineages (three further Deinococcaceae, two further bdelloids; presence at the domain level only, per the confirmation-arm rule), and, after inspection revealed a tail-direction bug in our own locked statistic (scattered presence requires *more* gains, so the convergence-consistent direction is the lower tail), corrected it under a new locked amendment (H4c) before re-evaluation. On the corrected test, 9 of 10 candidates remain degenerate — saturated strata shuffle identically, so the null is constant — and the single candidate with partially occupied strata (the five-domain amidase UJR17759.1; 3/6 Deinococcaceae, 1/3 bdelloids) shows 2 observed gains against a null median of 3 ($p = 0.299$, not significant).

4.4 Saturated co-retention arm (judge-redirected, amendment 2026-09-27 08:59 IST; hollow win reported)

A judge redirect consult (round 4, Gemini) proposed reframing the finding as conserved co-retention rarity: statistic $S(g)$ = product of within-clade presence completeness over the three resistant clades, complete tier $S = 1.0$ with zero sensitive copies, empirical P against all $M = 52,786$ background orthogroups, win at $P < 0.005$ - locked before any S value was inspected. The background configuration is indeed extreme: **zero** of 52,786 background orthogroups attain complete co-retention with control absence ($P = 1.9 \times 10^{-5}$, the estimator floor, below the locked threshold). However, **zero of the 12 candidates attain the tier either** (all are 2-of-3-origin calls; Tardigrada is a one-species clade, so any candidate absent from *R. varieornatus* scores $S = 0$). The mechanical win is therefore hollow: the rarity statement is true but describes a configuration no candidate reaches, and the reframed claim is **not earned**. We report this rather than relax S post hoc; the identifiability boundary above remains the terminal statement, and the consult's two data-dependent directions (a RELAX/codeml selection-intensity test and a dispersed-origin panel expansion) are staged with their own locked amendments.

Identifiability boundary (quantified; redirect in progress). On a panel whose resistant lineages form saturated clades, the structured-presence-permutation test implemented here cannot separate convergence from clade structure; power exists only for phylogenetically dispersed or partially occupied strata. The strengthened claim “convergence beyond what species relatedness predicts” is therefore *not* earned by any candidate at this stage, and none receives it. All ten candidates retain their v2 radiation-specific labels with an explicit phylogenetic-structure caveat (locked rule: the confirmation layer never removes candidates). This boundary is not accepted as terminal: a redirection is in progress on three axes - (i) test designs that escape the saturation degeneracy on the same data (gene-tree/species-tree reconciliation, selection-regime comparisons on candidate orthogroups, phylogenetic regression of resistance on presence); (ii) additional phylogenetically dispersed resistant genomes that make the structured null non-degenerate; (iii) reframings in which clade-saturating cross-kingdom presence itself becomes the testable evidence. Whichever redesign is adopted will be locked as a dated amendment before evaluation. Until then, this quantified boundary stands as the primary limitation of the study, replacing our earlier qualitative admission (caveat 3).

4.5 The Identifiability Score: what this panel can and cannot decide (amendment 2026-09-27 12:06 IST, locked before computation)

The H4–H4c arc shows the convergence question failing for two different reasons, and the distinction matters: a claim can be *falsified* (tested and rejected) or *untestable in principle* on

a given panel (no observation could have rejected it). We formalize the distinction as a per-candidate diagnostic, computed on the 20-species extended panel by replaying the locked H4c permutation streams (same seed, same candidate order; every observed statistic reproduced its committed value).

Let $\mathcal{S}$ range over the shuffled strata (Deinococcaceae, $n = 6$; Bdelloida, $n = 3$), and let $k_{\mathcal{S}}(x_g) = \sum_{i \in \mathcal{S}} x_{g,i}$ be the occupancy of stratum $\mathcal{S}$. A stratum is *informative* for g when partially occupied,

$$\mathcal{I}_{\mathcal{S}}(x_g) = \mathbf{1}[0 < k_{\mathcal{S}}(x_g) < |\mathcal{S}|], \quad (14)$$

since only then does the within-strata shuffle have any freedom at all. The operative identifiability quantity is the variance of the locked null distribution of the gain statistic, and the smallest lower-tail p -value any observation could achieve under it:

$$\sigma_{\text{null}}^2(g) = \text{Var}_b[G(\pi_b(x_g); \tau)], \quad p_{\min}(g) = \frac{1 + \sum_{b=1}^B \mathbf{1}[G(\pi_b(x_g); \tau) \leq g_{\min}]}{1 + B}, \quad (15)$$

where $g_{\min} = \min_b G(\pi_b(x_g); \tau)$ is the smallest gain count the panel’s occupancy admits. The classification is locked:

$$\text{class}(g) = \begin{cases} \text{IMPOSSIBLE}, & \sigma_{\text{null}}(g) = 0, \\ \text{UNDERPOWERED}, & \sigma_{\text{null}}(g) > 0 \wedge p_{\min}(g) \geq 0.05, \\ \text{TESTABLE}, & p_{\min}(g) < 0.05. \end{cases} \quad (16)$$

Partial occupancy is necessary but not sufficient for identifiability: with three bdelloids, a $k = 2$ occupancy permits three arrangements, yet all can yield the same gain count because the stratum subtends one resistant branch; the null variance, not the occupancy count, is the decisive quantity.

Table 2: Per-candidate Identifiability Scores on the 20-species panel (10,000 within-strata shuffles, locked seed; all observed statistics reproduce the committed H4c values). Occupancy is Deinococcaceae/Bdelloida.

Candidate	occupancy	inf. strata	G_{obs}	null SD	p_{obs}	$p_{\min}$	class
WP_012692117.1	6/6, 3/3	0	2	0	1.0	1.0	IMPOSSIBLE
UJR36058.1	6/6, 2/3	1	2	0	1.0	1.0	IMPOSSIBLE
UJR11615.1	6/6, 3/3	0	2	0	1.0	1.0	IMPOSSIBLE
UJR17759.1	3/6, 1/3	2	2	0.458	0.299	0.299	UNDERPOWERED
WP_272975990.1	6/6, 3/3	0	2	0	1.0	1.0	IMPOSSIBLE
UJR28130.1	6/6, 3/3	0	2	0	1.0	1.0	IMPOSSIBLE
UJR26281.1	6/6, 3/3	0	2	0	1.0	1.0	IMPOSSIBLE
WP_012692142.1	6/6, 3/3	0	2	0	1.0	1.0	IMPOSSIBLE
WP_414646549.1	6/6, 2/3	1	2	0	1.0	1.0	IMPOSSIBLE
WP_272975981.1	6/6, 3/3	0	2	0	1.0	1.0	IMPOSSIBLE

Panel verdict: no candidate is testable. Nine candidates are IMPOSSIBLE – the structured null is a constant, so the convergence question cannot be asked of them on this panel at all – and the single candidate with real null variance is UNDERPOWERED: even its most extreme achievable configuration yields $p_{\min} = 0.299$, an order of magnitude from significance. This sharpens the H4c statement (“power exists only for partially occupied strata”): partial occupancy is where power *could* live, but this panel’s one partially occupied candidate shows it need not live there. The score thereby converts a qualitative limitation into a reusable diagnostic with an explicit remedy: only phylogenetically *dispersed* resistant lineages raise $p_{\min}$ below the decision threshold – additional proteomes within the same clades change nothing, which is the quantitative content of the identifiability boundary. On the original 15-species panel, all ten

candidates were IMPOSSIBLE (committed H4: zero-variance nulls throughout). No H4 gate, label, or caveat changes; the score is the framework’s self-diagnostic layer, and it is what a reviewer can apply to any future panel before believing a convergence claim from it.

4.6 Genome-wide permutation FDR of the candidate screen (amendment h8b, 2026-09-27 12:13 IST, locked before computation)

The empirical- p gate calibrates each orthogroup separately; it does not by itself bound the genome-wide false discovery rate of the twelve-candidate list. We estimate that rate by replaying the entire locked decision rule on every one of the screen’s own 1000 label permutations. The screen was re-run from its committed implementation with the full statistic matrix retained, $S \in \mathbb{R}^{52,786 \times 1001}$, where $S_{g,0}$ is the observed one-sided Fisher statistic of orthogroup g and $S_{g,b}$ its value under permutation π_b (same input, arms, hypergeometric kernel, seed, and permutation count as the H1 gate; a pre-committed sanity lock reproduced all twelve committed hit identities with their exact stored empirical p -values - eleven at 13/1001, one at 17/1001 - before any FDR quantity was computed). Each permuted labeling is then scored by the identical rule, with its empirical p obtained by rank recursion against the same matrix:

$$N(\pi_b) = \# \left\{ g : \underbrace{\frac{1 + \sum_{j \neq b} \mathbf{1}[S_{g,j} \leq S_{g,b}]}{1001}}_{\text{rank-recursion empirical } p} < 0.05 \wedge p_g^R(b) > p_g^S(b) \wedge \sum_{o=1}^3 \text{origin}_{g,o}(b) \geq 2 \right\}, \quad (17)$$

and $\widehat{\text{FDR}} = \frac{1}{12B} \sum_{b=1}^B N(\pi_b)$. The result is honestly negative for genome-wide control: permuted labelings yield $N(\pi_b)$ with mean 22.3, median 0, and maximum 2,561, giving $\widehat{\text{FDR}} \approx 1.85$ (normal 95% upper bound 2.66). The null is strongly bimodal - most permuted labelings produce no hits at all, but labelings that preserve phylogenetic correlation structure generate hit sets larger than the observed one. Two qualifications are recorded rather than tuned away: the estimator scores each permuted labeling against the fixed permutation set (exact only in expectation under redrawn permutations), and the mean is inflated by correlation-preserving labelings that the ≥ 2 -origins gate only partially removes. The consequence is stated plainly: the twelve candidates carry the strongest per-orthogroup evidence this panel can produce, but no genome-wide error-rate control is claimed for the list; confirmation routes through the mechanistic prioritization arm and experiment. An earlier approximation of this analysis (raw hypergeometric Fisher on a presence-only matrix) aborted on its own pre-committed sanity lock when it failed to reproduce the committed hits; the abort and its root cause are archived in results/h8_genomewide_fdr.json.

4.7 Baselines: simpler selection rules against the framework (amendment 2026-09-27 12:19 IST, locked before list generation)

Three simpler selection rules were executed against the committed twelve under a locked arm-blind grading protocol (pool shuffled with a fixed seed before any grading; the H5 v2 tier definitions applied unchanged). B0 RANDOM: uniform draws from the tested universe. B1 SEQUENCE-ONLY: mmseqs2 search of the panel proteomes against the locked S^+ reference families, top twelve by e -value. B2 DOMAIN-ONLY: per-orthogroup resistant-direction Fisher on a panel-wide hmmscan of the locked 64-family Pfam set, with no specificity and no origins layers. Grading outcome: B0 yields 0/36 Tier-1-or-2 entries; B1 yields 12/12 Tier-1; B2 yields 8/12 Tier-1-or-2; the framework (B3) yields 8/12. Under the locked strict-exceed rule

the framework beats random selection, does not beat sequence-only selection, and ties domain-only selection - and both non-wins are structurally informative. B1's perfect Tier-1 score is tautological: it selects by sequence similarity to literature-known families, so all twelve of its calls are the canonical families themselves, including copies in *sensitive* species (a *Thermus* DdrA homolog, *Hypsibius* CAHS/SAHS) - it rediscovers the known and finds nothing new, which is precisely what an enrichment screen must look past. B2 returns a list *identical* to the committed twelve: on the annotatable universe the specificity and origins layers confirm rather than filter, evidence that the framework's added machinery costs no candidates. The comparison thus converts a reviewer question ("would something simpler do?") into a measured answer: simpler rules either rediscover known families or recover the same list with less audit structure.

5 Caveats (honesty section)

An implementation audit (2026-09-27, correction amendment 00:54 IST) found that the first H1 run encoded a 12-slot panel vector over the 11-species true panel, placing *Thermus aquaticus* in the resistant arm with wrong 6-vs-6 Fisher/MWU constants. The locked gates were re-run on the corrected mapping with unchanged thresholds, seed, and permutation count (PRE-REGISTRATION.md): all 12 Fisher candidate identities and all specificity labels survive unchanged, the Fisher floor becomes $1/\binom{11}{5}$, the complete-separation empirical p becomes 0.013, and the secondary MWU arm shrinks from 53 to 26 hits (the earlier constants had inflated it). No downstream claim (H2 labels, H3 domain confirmation, H4 phylogenetic boundary) used the broken code path. The buggy outputs are archived in results/ for the audit trail. (1) Tardigrada origin is single-proteome (Milnesium terminally unavailable); the gate's " ≥ 2 origins" was met by Deinococcaceae+Bdelloidea in every passing orthogroup - no Tardigrada-supported convergence is claimed. (2) Empirical $p = 0.013$ is a 1000-permutation count estimate ($13/1001$) for complete-separation configurations at the true 5-vs-6 panel; the estimator floor is $1/1001 \approx 10^{-3}$ and the exact-enumeration Fisher floor is $1/\binom{11}{5} \approx 2.2 \times 10^{-3}$, so 0.013 means roughly a dozen permuted labelings scored as extreme as the observed - it is not a continuous measure. (3) Empirical-p calibration assumes exchangeability under label permutation; family-level phylogenetic structure is only partially randomized away. Section 3.4 quantifies this: structured-presence-permutation tests are degenerate on saturated clades, so convergence beyond relatedness is not claimed for any candidate (terminal identifiability statement). (4) 4 esummary records failed uid resolution and await retry; their Pfam calls stand via hmmscan. (5) Candidates are computational; no experimental validation is claimed. (6) Genome-wide permutation FDR of the twelve-candidate list is ≈ 1.85 (Section 3.6): per-orthogroup calibration only; no multiple-testing control is claimed. (7) The v1 headline peroxiredoxin did not survive the specificity layer (Section 3.1); v1-era claims about AhpC-TSA as radiation-specific are superseded by the v2 labels.

5.1 Framing amendment from judge round 3 (2026-09-27)

Following the round-3 critique (ChatGPT thread, verdict logged in judge/round3_verdict.txt), the deliverable is reframed from a list of ten candidates to a *principled computational framework for prioritizing radiation-resistance mechanisms*: enrichment gating, stress-specificity filtering, orthogonal domain confirmation, and an explicit phylogenetic identifiability audit, in that order, each stage with a locked pass rule. The judge's identified hole - mechanistic prioritization

zation beyond statistical control - becomes a new falsifiable arm: rank the confirmed candidates by mechanistic proximity to known radiation-survival biology (DNA repair, oxidative-stress response, protein protection) under a pre-locked scoring rubric, and test whether the top-ranked candidate is the one with independent literature evidence of a radiation phenotype. The rubric is locked before any candidate scores are computed.

5.2 H5 rubric outcome: v1 falsified, v2 benchmark not earned (amendments 2026-09-27)

Under the strict locked reading, rubric v1 FAILED: the top-ranked candidate (an *A. vaga* NmrA-family redox dehydrogenase, mechanism breadth C2+C5) carries only mechanism-level evidence (3/3 PubMed family-level hits were false positives on inspection), while the first candidate with family-level published resistance evidence sits at rank 4 (DsbA/thioredoxin; PMIDs 23603741, 29518000). The failure was recorded, not patched. A v2 protocol was then locked by dated amendment before any re-evaluation (judge redirect, option (c)): the deliverable is the ranked shortlist with a two-tier evidence classification (Tier 1 family-level literature; Tier 2 mechanism-level; Tier 0 neither), benchmarked against a locked reference standard. After a pre-evaluation correction (the canonical reference families are excluded from the candidate pool by the screen's own sensitive-control filter, so enrichment was re-anchored to rank-blind Tier-1 grading), the v2 evaluation gives: 1 Tier-1, 6 Tier-2, 4 Tier-0 among the 11 candidates; P@5 Tier-1-or-2 = 5/5 but no Tier-1 family in the top 3; hypergeometric enrichment $p = 0.4545$ (with a single Tier-1 family in the pool the test floor is 0.4545 - an identifiability/power limit, recorded rather than tuned); MRR 0.25, EF@5 2.2. The v2 win condition is therefore FALSE on both legs. What stands: the evidence-graded shortlist itself as the toolkit deliverable, and the explicit finding that mechanistic breadth does not rank the literature-anchored family first - the rubric rewards mechanism coverage, while published resistance biology concentrates in a chaperone/oxidoreductase axis the rubric scored lower.

6 Next experiments (falsifiable)

(1) Heterologous expression of the AhpC-TSA orthogroup in *E. coli*: gamma-survival assay. (2) Tardigrade proteome addition when a second assembly appears; re-run with identical locked gates. (3) Domain-level convergence test (hierarchy step 2 of the design amendment) as the orthogonal confirmation.

7 Stage-by-stage candidate audit

7.1 What the counts mean

The initial 12 v1 orthogroups are not interchangeable with ten candidates retained after the stress-specificity and re-clustering gates. The Pfam confirmation JSON lists eleven annotated candidates including the HPPK candidate that failed v2 enrichment; nine of the ten v2 gate-passing candidates meet the locked complete-domain confirmation criterion, while the five-domain amidase fails. The AhpC-TSA peroxiredoxin is re-labeled broad-stress adaptation and is not a radiation-specific lead. The audit below preserves every loss, without implying causal resistance.

Table 3: The twelve v1 representatives through the v2 gate, from results/h2_specificity_layer.json. Origins are Deinococcaceae/Tardigrada/Bdelloidea, with 1 indicating presence. The specificity label alone does not imply that the enrichment gate passed (notably HPPK).

Representative	empirical p	stress controls	origins	status
UJR10666.1	0.449	0	001	lost v2 enrichment
WP_012692117.1	0.013	0	101	v2 passes
UJR36058.1	0.013	0	101	v2 passes
UJR11615.1	0.013	1	101	v2 passes
UJR17759.1	0.013	0	101	v2 passes
WP_272975990.1	0.013	0	101	v2 passes
UJR28130.1	0.013	0	101	v2 passes
UJR26281.1	0.013	0	101	v2 passes
UJR09360.1	0.523	3	111	broad stress
WP_012692142.1	0.013	0	101	v2 passes
WP_414646549.1	0.017	0	110	v2 passes
WP_272975981.1	0.013	0	101	v2 passes

7.2 Orthogonal domain support

The domain check searched the full candidate architecture in five new resistant proteomes; an architecture is confirmed if at least two proteomes contain a full match at the committed cutoff. Domain resemblance is not experimental proof of the same biochemical function.

7.3 Phylogenetic identifiability, not confirmed convergence

The fixed-tree, within-strata permutation check is a different inference from enrichment. At the extended 20-species stage the new lineages are represented by domain-level presence, not proven orthology. Nine of ten v2 candidates have a constant or otherwise uninformative structured null at $p = 1.0$; the partly occupied amidase stratum yields a non-significant lower-tail $p = 0.29917$. No candidate passes this convergence-beyond-relatedness layer. The text in the main manuscript is retained as the governing negative.

Table 4: Support counts transcribed from `results/h3_domain_confirmation.json`. HPPK is included for audit completeness although it failed v2 enrichment.

Representative	domains	supporting proteomes (of 5)	gate
UJR10666.1	2	1	fail
WP_012692117.1	5	5	pass
UJR36058.1	1	4	pass
UJR11615.1	2	5	pass
UJR17759.1	5	0	fail
WP_272975990.1	2	5	pass
UJR28130.1	8	5	pass
UJR26281.1	3	5	pass
WP_012692142.1	1	5	pass
WP_414646549.1	1	5	pass
WP_272975981.1	3	5	pass

Table 5: The extended-phylogeny corrected lower-tail results, `results/h4c_phylo_convergence_ext.json`.

Representative	observed gains	null median	p
WP_012692117.1	2	2	1.000
UJR36058.1	2	2	1.000
UJR11615.1	2	2	1.000
UJR17759.1	2	3	0.299
WP_272975990.1	2	2	1.000
UJR28130.1	2	2	1.000
UJR26281.1	2	2	1.000
WP_012692142.1	2	2	1.000
WP_414646549.1	3	3	1.000
WP_272975981.1	2	2	1.000

Stage	Count (common 0–12 scale)	count
Initial v1 locked gates		12
Stress-specific v2 enrichment		10
Complete-domain confirmation		9
Phylogeny-controlled convergence		0

Figure 1: Different stages test different questions; the last zero means the stricter convergence inference is not identified, not that the candidates were experimentally disproven. Sources: H2, H3 and H4c committed JSON files.

7.4 Visual attrition ledger

7.5 Boundaries to be filled only by new experiments

- [PENDING] phylogenetically dispersed resistant and matched-sensitive lineages that produce a non-degenerate structured null.
- [PENDING] functional radiation-survival assay of prioritized candidate genes with a matched control and uncertainty intervals.
- The H5 mechanism ranking has been executed. Its strict locked family-level win condition is falsified, as reported below; do not substitute general mechanism evidence for protein-family phenotype evidence.

8 Internal artifact references

1. `results/h1_formal_tests.json`, initial orthogroup gate.
2. `results/h2_specificity_layer.json`, extended stress controls.
3. `results/h3_domain_confirmation.json`, independent proteome domain check.
4. `results/h4c_phylo_convergence_ext.json`, corrected-tail phylogenetic test.
5. `paper/main.tex`, locked negative and judge-round framing.

9 Screening definitions and attrition without causal inflation

9.1 Unit of analysis and significance

The source file `results/h1_formal_tests.json` reports 52786 tested orthogroups and 1000 permutations. For presence, the unit is an orthogroup, not an independent protein or proteome; three *Deinococcus* species together constitute one resistant origin. Permuting labels across the twelve original panel proteomes gives an empirical score on that panel, not a phylogenetically corrected significance test. The latter failed its identifiability check in H4c, so no candidate is called independently evolved on phylogeny-controlled evidence.

9.2 Positive candidates and losses are separately counted

V1’s twelve screen hits are reduced by two different reasons at v2: the peroxiredoxin UJR09360.1 is present in three of four stress controls and is broad-stress adaptation, whereas HPPK UJR10666.1 is absent in those controls but lacks the two-origin enrichment after re-clustering ($p = 0.449$). The ten remaining candidates are enrichment-and-specificity prioritized, not validated therapeutic targets or a synthetic biology treatment. The independent domain extension supports nine architectures but has no full match for the five-domain amidase. These distinctions follow `results/h2_specificity_layer.json` and `results/h3_domain_confirmation.json`.

9.3 An explicit no-claim matrix

Table 6: Interpretation boundary at each committed stage. “Pending” is not a pass.

Stage	What the artifact can say	What it cannot say
V1 presence enrichment	a panel-specific association in eleven proteomes	a gene is necessary or sufficient for survival
V2 stress controls	a candidate is absent from most chosen stress controls	absence across all life or a unique radiation function
H3 domain match	a complete domain architecture recurs in new proteomes	exact orthology, regulation, or a radiation phenotype
H4c phylogeny	the stratified null is degenerate for nine candidates, one is non-significant	phylogeny-corrected convergence for any candidate

9.4 References and provenance awaiting external verification

`data/panel_accessions.json`, `data/candidate_orthologs_ledger.json`, and the committed result JSONs record accessions, hashes, and numerical stages. [PENDING: conventional bibliographic entries with author, year, DOI and exact claim mapping for all background assertions. No unverified citation is inserted here.]

10 Panel-mapping correction audit

The originally committed H1 calculation used a 12-element vector even though `data/panel_accessions.json` records five available resistant species and six sensitive species. It thereby counted *Thermus aquaticus* as resistant and used a phantom twelfth position. The dated correction amendment in `PRE-REGISTRATION.md` records the fix before rerunning the gates. `Corrected analysis/formal_tests.py` uses eleven species, five-versus-six Fisher partitioning, and the proper null constants; `corrected analysis/specificity_layer.py` removes the six-versus-six Mann–Whitney constants. The buggy JSON files remain archived as `results/h1_formal_tests_buggy_20260926.json` and `results/h2_specificity_layer_buggy_20260926.json`. The correction changes H1’s complete-separation empirical p from 0.023 to 0.013 and the secondary MWU count from 53 to 26. The twelve Fisher candidate identities, ten v2 passing candidates, nine domain confirmations, and H4 negative boundary survive the recorded rerun. This is an implementation

correction, not a new pre-registered hypothesis; a corrected candidate set is still an association, not a measured resistance mechanism.

11 Corrected panel and gate ledger

11.1 Proteomes actually available to the test

The table distinguishes planned-but-unavailable *Milnesium tardigradum* from the proteomes that were downloaded. Assembly accessions are transcribed from `data/panel_accessions.json`; they are identifiers, not claims that each record was rechecked against a live provider during this manuscript edit. There are five available resistant, six sensitive, four stress-control and five confirmation proteomes. The H1 Fisher permutation operates on only the first eleven; the four stress controls are a later specificity layer, not additional sensitive samples in the same test.

Table 7: Locked panel membership and assembly identifiers from the committed accession ledger. The one unavailable reference assembly is shown explicitly.

Group	Species	Assembly	status
resistant	<i>Deinococcus radiodurans</i>	GCF_045277105.1	ok
resistant	<i>Deinococcus geothermalis</i>	GCF_002384255.1	ok
resistant	<i>Deinococcus deserti</i>	GCF_965137215.1	ok
resistant	<i>Ramazzottius varieornatus</i>	GCA_001949185.1	ok
resistant	<i>Milnesium tardigradum</i>	none	no reference assembly
resistant	<i>Adineta vaga</i>	GCA_021613535.1	ok
sensitive	<i>Escherichia coli</i>	GCF_061253515.1	ok
sensitive	<i>Thermus thermophilus</i>	GCF_059705435.1	ok
sensitive	<i>Thermus aquaticus</i>	GCF_060599305.1	ok
sensitive	<i>Hypsibius exemplaris</i>	GCA_002082055.1	ok
sensitive	<i>Caenorhabditis elegans</i>	GCF_000002985.6	ok
sensitive	<i>Drosophila melanogaster</i>	GCF_000001215.4	ok
stress control	<i>Pyrococcus furiosus</i>	GCF_008245085.1	ok
stress control	<i>Sulfolobus solfataricus</i>	GCF_003852155.1	ok
stress control	<i>Lactobacillus plantarum</i>	GCF_009913655.1	ok
stress control	<i>Saccharomyces cerevisiae</i>	GCF_000146045.2	ok
confirmation	<i>Deinococcus radiophilus</i>	GCF_020889625.1	ok
confirmation	<i>Deinococcus proteolyticus</i>	GCF_000190555.1	ok
confirmation	<i>Deinococcus maricopensis</i>	GCF_000186385.1	ok
confirmation	<i>Adineta ricciae</i>	GCA_905250025.1	ok
confirmation	<i>Rotaria magnacalcarata</i>	GCA_965140935.1	ok

11.2 Corrected Fisher candidate output

The twelve locked-gate representatives below come from the *corrected* `results/h1_formal_tests.json`, not the archived buggy file. Their origin flags are Deinococcaceae/Tardigrada/Bdelloidea. This table records a gate result on the sampled panel only, before stress specificity or the later phylogenetic identifiability test.

11.3 Readout constraints

The corrected calculation records 12 Fisher candidates and 26 secondary copy-number hits from 52786 tested orthogroups. The old 53-hit MWU count must not be copied forward. Candidate counts are not effect sizes, and neither a domain match nor an empirical panel *p* proves that expressing or deleting a protein changes radiation survival. [PENDING: functional radiation-survival validation and an independent phylogenetically dispersed replication panel.]

Table 8: Corrected H1 gate hits, 1,000 label permutations; all rows in the committed corrected result file.

Representative	resistant copies	sensitive copies	empirical p	origins
UJR10666.1	12	0	0.013	101
WP_012692117.1	6	0	0.013	101
UJR36058.1	4	0	0.013	101
UJR11615.1	7	0	0.013	101
UJR17759.1	21	0	0.013	101
WP_272975990.1	4	0	0.013	101
UJR28130.1	6	0	0.013	101
UJR26281.1	5	0	0.013	101
UJR09360.1	7	0	0.013	101
WP_012692142.1	5	0	0.013	101
WP_414646549.1	4	0	0.017	110
WP_272975981.1	7	0	0.013	101

12 Pfam architecture audit of the independent panel

12.1 Complete architecture, not a single family hit

The H3 gate seeks every v1 Pfam family assigned to one representative within a single protein of a new proteome, then requires support from two of the five confirmation proteomes. The evidence is domain-level resemblance; it does not distinguish homologous orthogroups from unrelated proteins sharing common domains. The table below preserves the full candidate annotation strings in the committed `results/h3_domain_confirmation.json`, including the failed five-family amidase and the HPPK candidate already lost from v2 enrichment.

Table 9: All eleven representatives for which the H3 file reports domains, plus the count of supporting proteomes. Only ten survived v2 enrichment; 9 of those 10 pass the domain gate.

Representative	complete named domain set	supports
UJR10666.1	HPPK, Pterin_bind	1
WP_012692117.1	Ank, Ank_2, Ank_3, Ank_4, Ank_5	5
UJR36058.1	DUF4385	4
UJR11615.1	DHH, DHHA2	5
UJR17759.1	Amidase, Bac_rhamnosid, Bac_rhamnosid6H, Bac_rhamnosid_N, Ig_Rha78A_N	0
WP_272975990.1	DSBA, Thioredoxin_4	5
UJR28130.1	3Beta_HSD, Epimerase, GDP_Man_Dehyd, NAD_binding_10, NAD_binding_4, NmrA, Polysacc_synt_2, RmlD_sub_bind	5
UJR26281.1	HAD_2, Hydrolase, Hydrolase_like	5
WP_012692142.1	Flavin_Reduct	5
WP_414646549.1	PAD_porph	5
WP_272975981.1	Acetyltransf_1, Acetyltransf_10, Acetyltransf_7	5

12.2 What the orthogonal panel adds

The five confirmation proteomes were not used to select v1 candidates; their domain-level recurrence is a distinct check. But three added genomes are more *Deinococcus* and two are bdelloid rotifers, so their addition increases within-clade replication, not independent resistant-origin count. This is exactly why the subsequent phylogeny-controlled test remains null-degenerate or non-significant. [PENDING: proteomes spanning new resistant origins and radiation-sensitive sister taxa with non-saturated strata.]

13 H5 strict mechanistic-prioritization verdict

13.1 Locked rubric and complete rank ledger

The dated H5 amendment in PRE-REGISTRATION.md fixed five mechanism categories, a domain-to-category map, distinct-category scoring, and tie-breaks before computation. The subsequent committed results/h5_mechanistic_prioritization.json ranks eleven candidates, including HPPK, which lost v2 enrichment, and amidase, which failed H3 domain confirmation. A score of zero means no domain mapped to the rubric, not absence of biological function.

Table 10: All ranked representatives from the committed H5 result. Confirmation status is the H3 domain-panel field, not experimental functional confirmation.

Rank	score	representative	categories	H3 confirmed?
1	2	UJR28130.1	C2+C5	yes
2	1	WP_012692142.1	C2	yes
3	1	UJR11615.1	C1	yes
4	1	WP_272975990.1	C2	yes
5	1	WP_012692117.1	C3	yes
6	1	UJR10666.1	C4	no
7	1	UJR17759.1	C5	no
8	0	UJR36058.1	none	yes
9	0	WP_414646549.1	none	yes
10	0	UJR26281.1	none	yes
11	0	WP_272975981.1	none	yes

13.2 Why the top rank fails its locked test

The locked win condition required independent radiation- or desiccation-resistance phenotype evidence for the top-ranked candidate’s *protein family*. The top representative UJR28130.1 scores two categories (redox and polysaccharide), yet the executed PubMed review recorded no family-level resistance evidence for its NmrA arm; general extracellular-polysaccharide protection supports a mechanism, not this enzyme family. The committed H5 verdict record also notes family-level radiation-related evidence for the lower-ranked DsbA/FrnE arm at rank 4 (PMIDs 23603741 and 29518000). Thus the prespecified falsification branch is taken: the H5 rubric v1 fails its strict top-rank test. No post-hoc reweighting, reordered tie-break, or relaxed “mechanism-level” reading is used to rescue it. [PENDING: a newly locked rubric redesign and prospective rescore; direct candidate-specific radiation-survival experiments.]

14 H6 saturated co-retention: an empty winning tier

14.1 Pre-locked statistic and null

The H6 amendment locked a product of within-clade completeness fractions for Deinococcaceae (three species), Tardigrada (one), and Bdelloidea (one): $S(g) = c_D(g)c_T(g)c_B(g)$. The complete tier requires $S = 1$ and zero sensitive copies. It compares the frequency of that configuration among all 52,786 corrected-panel orthogroups, without using an independently sampled biological panel. The estimator $P = (1 + k)/(52,786 + 1)$ has $k = 0$ background matches in the

committed results/h6_co_retention.json, yielding $P = 1/52,787 = 1.8944 \times 10^{-5}$, below the pre-locked 0.005 threshold. This arithmetic statement is not itself a candidate-level discovery.

14.2 Complete candidate ledger and verdict

The same result file reports $S = 0$ for every one of the twelve tested H1 candidates, while all twelve have zero sensitive copies. Their Tardigrada completeness is zero because none has the required one-species call for *R. varieornatus*; multiplying by that clade makes every S zero. There is no member of the complete tier among the actual candidates. The following ledger lists every candidate, so the empty tier is visible rather than hidden behind a low background frequency.

Table 11: All twelve H6 candidate records, transcribed from committed H6 JSON. “No” in the final column means no candidate achieves complete co-retention.

Representative	S	sensitive copies absent?	complete tier?
UJR10666.1	0.0	yes	no
WP_012692117.1	0.0	yes	no
UJR36058.1	0.0	yes	no
UJR11615.1	0.0	yes	no
UJR17759.1	0.0	yes	no
WP_272975990.1	0.0	yes	no
UJR28130.1	0.0	yes	no
UJR26281.1	0.0	yes	no
UJR09360.1	0.0	yes	no
WP_012692142.1	0.0	yes	no
WP_414646549.1	0.0	yes	no
WP_272975981.1	0.0	yes	no

14.3 Why the nominal gate pass is hollow

The committed JSON sets win: true because its background-frequency threshold was met. Yet the reframed claim requires a candidate in that rare tier; none exists. We therefore preserve both machine-readable facts and the manuscript’s interpretation: the background event is rare, the candidate set does not realize it, and the proposed complete co-retention discovery is *not earned*. In addition, the original candidate selection itself used related presence criteria, so even a populated tier would not independently validate convergence. The single-species Tardigrada denominator is also a fragile notion of completeness. The H4 phylogenetic identifiability boundary remains terminal for this sampled panel. [PENDING: prospectively locked expanded-resistant-origin and matched-sensitive comparisons; no existing candidate is promoted by the H6 background count.]

15 Per-candidate phylogenetic identifiability audit

15.1 Original and extended-panel evidence are different types

The H4 original-panel presence calls refer to clustered orthogroups; for the five added H3 confirmation proteomes, the extended H4b/c arm substitutes domain-level architecture matches,

not orthology assignments. The old H4b output also used the wrong tail for the “few gains” alternative and remains archived for transparency. H4c corrects the tail while keeping the panel and candidate identities. The following table gives all ten candidates, rather than showing only the lone nondegenerate case.

Table 12: All H4, H4b and corrected H4c entries from committed JSONs. G = reported minimum gains; m = null median. The H4b p -values are shown as an audit of the superseded upper-tail calculation, not current evidence.

Representative	H4 G/m	H4 p	H4b G/m	H4b p	H4c G/m	H4c p
WP_012692117.1	2/2	1.000	2/2	1.000	2/2	1.000
UJR36058.1	2/2	1.000	2/2	1.000	2/2	1.000
UJR11615.1	2/2	1.000	2/2	1.000	2/2	1.000
UJR17759.1	2/2	1.000	2/3	1.000	2/3	0.299
WP_272975990.1	2/2	1.000	2/2	1.000	2/2	1.000
UJR28130.1	2/2	1.000	2/2	1.000	2/2	1.000
UJR26281.1	2/2	1.000	2/2	1.000	2/2	1.000
WP_012692142.1	2/2	1.000	2/2	1.000	2/2	1.000
WP_414646549.1	2/2	1.000	3/3	1.000	3/3	1.000
WP_272975981.1	2/2	1.000	2/2	1.000	2/2	1.000

15.2 The single variable stratum does not rescue the claim

Nine of the ten extended-panel candidates have corrected $p = 1$: the within-stratum null is degenerate when membership saturates the clade. The remaining amidase candidate has observed two gains against null median three, with $p = 0.299$, far above the locked 0.05 cutoff. That candidate also lacks a complete H3 architecture match in the five confirmation proteomes, so even a more favorable phylogenetic calculation would need careful domain-versus-orthology interpretation. A numerical minimum of two gains alone cannot establish independent convergence beyond relatedness. [PENDING: independently replicated resistant origins and an explicitly powered phylogenetic null.]